Novel Distribution Distance Based on Inconsistent Adaptive Region for Change Detection Using Hyperspectral Remote Sensing Images

Zhiyong Lv, *Senior Member, IEEE*, Zhengjie Lei, Linfu Xie, Nicola Falco, *Senior Member, IEEE*, Cheng Shi, *Member, IEEE*, and Zhenzhen You, *Member, IEEE*

***Abstract*— Change detection with remote sensing images (RSIs) plays an important role in the community of remote sensing applications. However, when change detection is conducted with hyperspectral RSIs (HRSIs), how to measure the change magnitude between bitemporal HRSIs becomes challenging due to the high dimension of HRSIs. In this article, a novel distribution distance based on inconsistent adaptive region (D^2IAR) change detection approach is proposed to measure the change magnitude between bitemporal HRSIs for improving the performance of change detection with HRSIs. First, a band selection algorithm called optimal neighborhood reconstruction (ONR) is employed to reduce the dimensions of HRSIs. Then, an adaptive region around each pixel is generated to explore the contextual feature around each pixel, and kernel density estimation (KDE) is suggested to estimate the spectral distribution of the pixels within an adaptive region. A distribution distance is defined based on the adaptive region to measure the change magnitude between bitemporal HRSIs. Finally, the change magnitude between pairwise adaptive regions is measured by the proposed distance between the pairwise distributions. Experimental results based on four datasets and comparisons with eight methods indicated the feasibility and superiorities of the proposed D^2IAR-based change detection approach with HRSIs. The improvement rates are approximately 0.13%–24.04% for overall accuracy. The code and datasets can be available at: https://github.com/ImgSciGroup/2024-HSICD.**

***Index Terms*— Hyperspectral remote sensing images (HRSIs), land cover change detection (LCCD), nonsubsampled shearlet transform (NSST), spatial context.**

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Zhiyong Lv is with the School of Computer Science and Engineering, Xi'an University of Technology, Xi'an 710048, China, and also with the Key Laboratory of Urban Land Resources Monitoring and Simulation, Ministry of Natural Resources, Shenzhen 518034, China (e-mail: Lvzhiyong_fly@hotmail.com).

Zhengjie Lei, Cheng Shi, and Zhenzhen You are with the School of Computer Science and Engineering, Xi'an University of Technology, Xi'an 710048, China (e-mail: leizhengjie_csz@outlook.com; zhenzhen_you89@163.com; chengc_s@hotmail.com).

Linfu Xie is with the State Key Laboratory of Subtropical Building and Urban Science and the Research Institute for Smart Cities, School of Architecture and Urban Planning, Shenzhen University, Shenzhen 518060, China, also with the Shenzhen Key Laboratory Digital Twin Technology Cities, Shenzhen 518060, China, and also with the Guangdong Hong Kong Macau Joint Laboratory Smart Cities, Shenzhen 518060, China (e-mail: linfuxie@szu.edu.cn).

Nicola Falco is with the Climate and Ecosystem Sciences Division, Lawrence Berkeley National Laboratory, Berkeley, CA 94720 USA (e-mail: nicolafalco@lbl.gov).

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I. Introduction

LAND cover change detection (LCCD) is defined as the identification of land cover change on the Earth's surface by comparing bitemporal remote sensing images (RSIs) covering the same geographic area but are acquired at different dates [1], [2]. In the past decades, LCCD with RSIs has been widely used in land cover database updating [3], natural disaster monitoring [4], environmental management [5], and agriculture and forestry monitoring [6]. Particularly, hyperspectral RSIs (HRSI) are an important data source in the community of LCCD with RSIs [7]. Compared to other types of RSIs, the advantage of HRSIs lies in the more than hundreds of spectral bands they contain, enabling them to provide richer spectral information. However, the advantage of HRSIs in change detection tasks has always caused some challenges, which can be summarized briefly as follows:

1) Compared with the other types of RSIs, more pseudo changes may be caused by the difference imaging condition of bitemporal HRSIs [8], [9]. Because the different external conditions, such as different atmospheric conditions, soil moisture, and sun heights, will create more complex change in a hyperfeature space than that in a lower feature space. These factors always result in pseudo changes in a detection map. In particular, the bands of remote sensing increase, and the external imaging condition difference may cause more pseudo changes for LCCD with HRSIs.

2) Measuring the change magnitude between bitemporal HRSIs for LCCD is challenging. Aside from the challenges in the community of LCCD with RSIs, due to the spectral and spatial complexity of HRSIs, how to measure the change magnitude between pairwise HRSIs is a pivotal challenge for the task of LCCD.

3) Utilizing spatial-contextual information via adaptive region is challenging for LCCD with HRSIs [10], [11].

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Intuitively, a large number of studies have demonstrated that exploring spatial contextual information via adaptive regions for RSIs can improve application performance and accuracy. However, the inconsistency of pairwise adaptive regions from bitemporal HRSIs in terms of shape and size caused a conflict in measuring change magnitude.

In the past decades, various methods have been proposed to overcome the challenges caused by HRSIs for LCCD [12], [13]. In this article, these methods can be categorized into two classes: traditional and deep learning. Details of these two methods are reviewed below.

The first widely used and basic traditional method is the change vector analysis (CVA)-based approach. The CVA-based LCCD approach concentrates on performing algebraic operations on each pixel to reflect the difference between bitemporal HRSIs, such as compressed CVA (C^2VA) [14], sequential-spectral CVA (S^2CVA) [15], and robust CVA [16]. In these approaches, all the spectral bands of inputting images are adopted to construct a vector, and the Euclidean distance is usually used to measure the change magnitude between the pairwise vectors from bitemporal RSIs. Compared with image difference methods, the CVA-based approach can achieve better performance with higher detection accuracies [17], [18]. However, employing all the bands to measure change magnitude may also bring noise to a detection map. Thus, CVA-based approaches are not suitable for applying to HRSIs in the LCCD community. Transformation-based approaches are also attractive in LCCD with HRSIs. These methods focus on transforming HRSIs to the same feature space where bitemporal HRSIs can be comparable. The typical studies of transformation-based methods are multivariate alteration detection (MAD) [19], iteratively reweighted MAD (IR-MAD) [20], slow feature analysis (SFA) [21], and principal component analysis (PCA)-based methods [22], [23]. Specifically, these methods aim to project the original data into a transformed space that can highlight the separateness of change and unchanged features. Various studies have indicated that although the method based on image transformation can effectively remove data redundancy and eliminate the interference of noise to a certain extent, it ignores spectral continuity and damages the similarity between adjacent pixels [24]. Besides the above methods, LCCD with HRSIs can be achieved by comparing the classification maps from bitemporal HRSIs [25]. Post-classification change detection methods conducted classification operations on the bitemporal HRSIs and then compared the classified map [26]. Therefore, these methods have an advantage in obtaining a "from-to" change detection map. However, the accuracy of change detection is over-dependent on the classification results [50]. Apart from the above methods, because HRSIs are acquired into a 3-D cube with multiple spectral bands, tensor expression is also a common technique to analyze HRSIs from the spatial and spectral aspects [27], [28]. Moreover, features obtained by decomposing and reconstruction tensors from different directions tend to get information about more dimensions and reduce spectral noise very well. Overall, traditional change detection methods aim at obtaining detection maps without predefined samples, the acquisition of which is considered time-consuming and labor-intensive. However, the performance of many existing traditional methods is too complex to be applied widely in practical applications, and some of these methods require empirical parameters to achieve satisfactory change detection performance [29].

In recent years, deep-learning neural networks have found widespread and successful applications in natural language processing (NLP) [30] and computer vision (CV) [31]. The excellent performance of these neural networks in practical applications is due to their capacity for exploring deep and abstract features [32]. The development of many hyperspectral change detection (HCD) algorithms has also been proposed [33], [34]. One of the typical deep learning networks is convolutional neural networks (CNNs), which has an advantage in exploring deep and abstract features from the inputting dataset [35]. For example, Wang et al. [36] proposed a universal end-to-end 2-D CNN named general end-to-end 2-D CNN framework (GETNET) to complete rich feature representation from a pixel-level mixing matrix for detecting change. Zhan et al. decomposed the change tensor along the spectral direction and the two spatial directions, then sent the extracted features to a CNN framework to accomplish the HCD process [37]. In addition, attention mechanism-based networks have an advantage in concentrating on the targets and obtaining satisfactory change detection performance. Hu et al., for instance, proposed a new pixel-level self-supervised hyperspectral spatial-spectral feature understanding network (HyperNet) to explore the spatial correlation of HRSIs [38]. Compared with traditional methods, deep learning models can learn different features from different layers and handle high-dimensional problems. However, although deep learning-based methods can achieve good detection accuracy, they are all data-driven, and the training of the network requires a large number of samples, the collection of which is time-consuming and labor-intensive.

Along with the many traditional methods, various studies indicated that utilizing spatial-contextual information around a pixel in a deep learning neural network is useful for improving change detection performance [39], [40]. One of the widely used methods for utilizing spatial-contextual information is a sliding window in the community of LCCD with RSIs [41]. For example, Celik [42] combined PCA and k-means clustering (PCA-K-Means) with a sliding window to analyze change magnitude. Many deep-learning networks have also been built on the utilization of regular image blocks to cover neighboring information, such as typical deep-learning structures 2D-CNN [43], [44] and U-Net [45], [46]. To cover more ground targets with various shapes and sizes, multiscale feature extraction methods [47], multiscale feature fusions [48], and hierarchy-scale analysis [12] have been proposed for LCCD with RSIs. However, even though utilizing the neighboring contextual information in a multiscale manner can achieve improved change detection performance compared to using a single scale, extending the scales in the process of feature extraction with different kernels and regular windows is insufficient to cover the ground targets with various shapes and sizes. To deal adaptively with targets of various shapes and sizes, adaptive region-based methods have become attractive

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in the application of LCCD with RSIs [10], [11]. All these approaches assume that the pixels from bitemporal images share the same adaptive region in terms of shape and size, and then measure the change magnitude between the pixels by using the pixels within pairwise adaptive regions from the bitemporal images. Intuitively, however, the pairwise adaptive regions generated around one pixel in bitemporal images may be different because bitemporal images may be acquired at different external imaging conditions, such as different sun heights and phenological seasons [49], [50].

In this article, with the motivation of utilizing spatial-contextual information, we proposed a novel change detection approach based on adaptive regions around each pixel to improve the performance of LCCD with HRSIs. The major and detailed contributions of the proposed approach can be briefly summarized as follows.

1) A novel metric distance named D²IAR-based change detection approach is first proposed to measure the change magnitude between bitemporal HRSIs. How to measure the change magnitude between bitemporal images is well known to be pivotal to obtaining satisfactory change detection performance. Therefore, proposing a novel distance metric approach is significant and may provide some potential ability for improving the detection performance of the LCCD with HRSIs.

2) A change magnitude image (CMI) fusion approach based on a nonsubsampled shearlet is first developed with the aim of converting the CMIs from each pairwise band of HRSIs into the final CMI for LCCD with HRSIs. Intuitively, one of the challenges in the community of LCCD with HRSIs lies in how to fuse the CMI between each pairwise band into a final CMI that describes the ground target change of the bitemporal HRSIs. Therefore, proposing a CMI fusion approach is beneficial to extend the application of LCCD with HRSIs.

3) Advanced change detection accuracies can be achieved by our proposed approach. Based on the proposed D²IAR-based change detection approach and the CMI fusion approach, some comparative experiments with four pairs of HRSIs have verified the feasibility and advantages of the proposed change detection approach. Achieving advanced detection accuracies in an unsupervised manner using the proposed approach may be preferable in practical applications.

The remaining parts of this article are organized as follows. The details of the proposed method are discussed in Section II. Experimental findings are presented in Section III, and the conclusion is provided in Section IV.

II. Methodology

The proposed D²IAR-based change detection approach with HRSIs aims to utilize spatial-contextual information in an adaptive manner to improve the change detection performance with HRSIs. In this section, the proposed approach will be introduced from two perspectives, including an overview of the proposed method framework, adaptive region generation and distribution distance, fusion of CMIs to a CMI based on nonsubsampled shearlet transform (NSST) [51], and the generation of a binary change map based on a Markov random field (MRF) model [52]. Each part will be detailed in Fig. 1.

A. Overview of the Proposed D²IAR-Based Change Detection Approach

As shown in Fig. 1, a band-selection algorithm named optimal neighborhood reconstruction (ONR) is first adopted to select three bands from the bitemporal HRSIs as the input feature for the change detection task [53]. Second, an adaptive region around each pixel in each band is generated to utilize spatial-contextual information. Next, the probability density distribution of each pairwise adaptive region is calculated to measure the change magnitude between pairwise pixels instead of directly using the spectral value of the pixel. Finally, after the entire images are scanned and processed with the adaptive region and the corresponding probability density distribution, the CMI between each pairwise band of the bitemporal HRSIs can be generated, and an MRF is employed to divide CMI into a binary change detection map (BCDM).

B. Proposed Adaptive Distribution Distance

Various studies have indicated that generating an adaptive region around each pixel is a useful way to utilize and explore spatial-contextual information for improving the application performance of RSIs [54]. However, adaptive regions around the pixels at the same position from bitemporal HRSIs are usually inconsistent in terms of shape and size, and how to compare the similarity between pairwise adaptive regions to measure the change magnitude using the pixels directly is a challenge. Therefore, a novel Distribution Distance based on Inconsistent Adaptive Region (D²IAR) change detection approach is proposed to measure the change magnitude between pairwise pixels from bitemporal HRSIs. Here, the adaptive region generation approach is borrowed from our previous study [2], [11], and the focus is on how to measure the distribution similarity between pairwise adaptive regions to achieve the CMIs. Details are presented below. First, a kernel density estimation (KDE) function is adopted to estimate the probability density distribution of the pixels within an adaptive region. Particularly, the Gaussian kernel is adopted as the kernel function because the spectral distribution of the pixels within an adaptive region usually meets a Gaussian normal distribution [55]. The formula is given in the following equation:

$$\hat{\rho}_{i,j}(x) = \frac{1}{\sqrt{2\pi n h}} \sum_{i=1}^{n} \exp\left(-\frac{(x - x_k)}{h^2}\right) \tag{1}$$

where $\hat{\rho}_{i,j}(x)$ is the probability density function based on the adaptive region generated around a pixel at the position (i,j), n is the number of pixels within the adaptive region, and h is a smoothing parameter called the bandwidth. Intuitively, a smaller h means a more detailed description of a distribution. However, there is always a trade-off between the bias of the estimator and its variance. In our proposed approach, σ and n respectively represent the standard deviation and number of pixels in the adaptive region. In particular, when estimating

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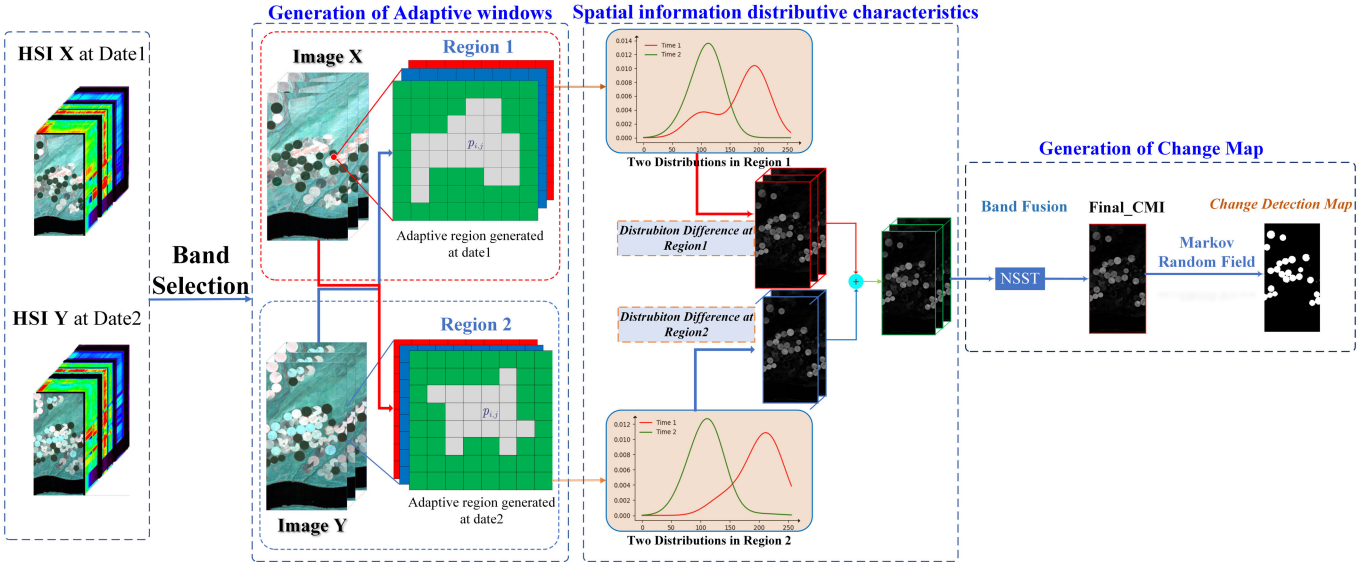


Fig. 1. Flowchart of the proposed D^{2}IAR-based change detection approach with HRSIs.

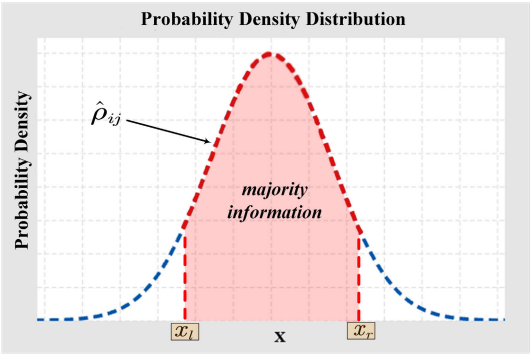


Fig. 2. Probability density distribution of the pixels within an adaptive region.

the probability density of a pixel (x) in a probability density function, it must be combined with that of other pixels (x_k) in terms of spectral value.

After obtaining the probability density function of an adaptive region, we discard some information on both sides of the distribution. This is done because the minority pixels with lower probabilities may be identified as spectral noise in the adaptive region. After we estimate the probability density functions respectively for the adaptive regions at the position (i,j) in the bitemporal HRSIs, where the probability density functions are denoted as $\hat{\rho}_{ij}^1$ and $\hat{\rho}_{ij}^2$, then the change magnitude between the pairwise adaptive regions can be defined by the difference between pairwise expectations. Here, the value of the expectation describes the majority information of the pixels within the adaptive region, as shown in Fig. 2. The formula for this process is shown in the following equation:

$$d_{i,j} = \left\| \int_{x_{l_1}}^{x_{r_1}} x\hat{\rho}_{ij}^1(x)dx - \int_{x_{l_2}}^{x_{r_2}} x\hat{\rho}_{ij}^2(x)dx \right\| \tag{2}$$

where $d_{i,j}$ denotes the distance between two expectations, and x_{l_1} and x_{l_2} are value domains of the probability density function. However, the adaptive regions around pixels (i,j) in bitemporal HRSIs usually vary in shape and size due to the differences in their atmospheric conditions and seasonal phenologies. To solve the problem of inconsistency, a simple yet effective approach is proposed to calculate the average distance between distributions. The equation consists of two parts as follows:

$$D_{i,j} = \left\| \frac{1}{2}\left(d_{i,j}^{I_1} + d_{i,j}^{I_2}\right) \right\| \tag{3}$$

where $d_{i,j}^{I_1}$ is the distance between the pairwise pixels at position (i,j) by using the adaptive region generated around the pixel at the position (i,j) in the preevent image (I_1), and $d_{i,j}^{I_2}$ is calculated based on the adaptive region generated around the pixel at the post-event image (I_2). Thus, $D_{i,j}$ is the change magnitude between the pairwise pixels at the position (i,j) in the bitemporal HRSIs. A larger $D_{i,j}$ value means a higher change probability between the pairwise pixels, whereas a smaller $D_{i,j}$ value implies a lower change probability between the pairwise pixels.

To demonstrate the feasibility of using probability density distribution in measuring the change magnitude between two pairwise adaptive regions around a pixel, an example of changed and unchanged areas is presented in Fig. 2. where P_{i_1,j_1} and P'_{i_1,j_1} denote the pairwise pixels (i_1, j_1) located in an unchanged area. The probability density distribution in Fig. 3 indicates that the majority of pixels within the pairwise adaptive region around the pixels (i_1, j_1) are distributed in the spectral value range from 100 to 150. However, when we observe the probability density distribution of the pixels within the adaptive region around the pixels (P_{i_2,j_2} and P'_{i_2,j_2}) in the changed area, the majority of pixels within the adaptive region around the pixel (P_{i_2,j_2}) in the preevent image distributed in the spectral value range from 50 to 100, whereas the majority of pixels within the adaptive region around the pixel (P'_{i_2,j_2}) in the post-event images distributed in the spectral value range from 150 to 200. Therefore, measuring the change magnitude between the bitemporal images via probability density distribution distance is feasible.

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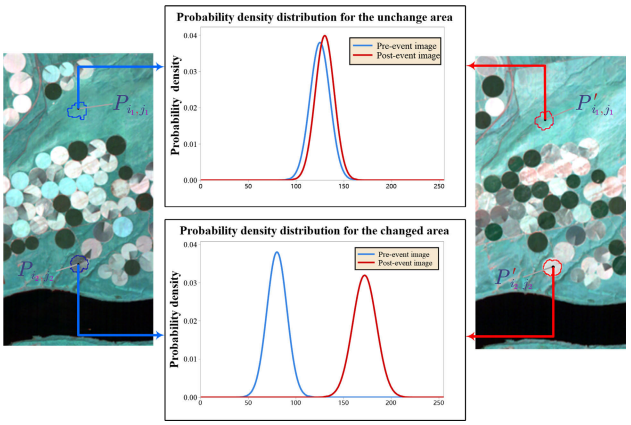


Fig. 3. Example of the probability density distributions between the pixels in changed and unchanged areas.

C. Fusing CMIs to a CMI With NSST

After generating the CMIs between each pair of bands, how to fuse the CMIs into a single CMI for achieving the final BCDM remains a problem. Here, a multiscale transformation approach named NSST was introduced to fuse CMIs to a CMI. Details of the operations are summarized below. First, NSST is applied to each CMI, which aims to decompose one image into two parts: low frequency and high frequency. Compared with the widely and commonly used wavelet transform, NSST can decompose an image from multiple directions to obtain low-frequency and high-frequency change information. Then, the corresponding low-frequency and high-frequency information in each CMI are fused to obtain the final CMI. Since the low-frequency coefficient reflects the overall information of the image, the high-frequency coefficient reflects the detailed information of an image. Therefore, the average of all low-frequency coefficients in all CMIs is considered the corresponding low-frequency coefficient of the final CMI, and the maximum value in the high-frequency coefficient in all CMIs is equal to the high-frequency coefficient. The change areas in a CMI usually perform as high-frequency information, and selecting a maximum value in the high frequency helps distinguish the change area in the final CMI. Finally, the fused CMI is obtained through inverse change reconstruction after fusing the high- and low-frequency information of multiple CMIs. The fusion details of the corresponding CMIs are shown in the following equation:

$$\begin{cases} \mathrm{CMI}_{\mathrm{Low}}(i,j) = \dfrac{1}{3}\displaystyle\sum_{b} \mathrm{CMI}^{b}_{\mathrm{Low}}(i,j) \\ \mathrm{CMI}_{\mathrm{High}}(i,j) = \max\left(\mathrm{CMI}^{b}_{\mathrm{High}}(i,j)\right) \end{cases} \quad b = 1, 2, 3. \tag{4}$$

D. Obtaining the Binary Change Map by MRF

In many existing studies of change detection with RSIs, each pixel in a CMI needs to be classified into a changed or unchanged label. In this work, an MRF model is selected to obtain the final change map. According to MRF theory, the problem of identifying whether a pixel has changed is transformed into the energy minimization problem of this pixel [56]. If the energy of a pixel in the change field is lower than its energy in the unchanged field, then it can be identified as changed; otherwise, it will be identified as unchanged. Based on this rule, the final change map is obtained by classifying each pixel in the MRF.

TABLE I
DETAIL DESCRIPTION FOR EACH DATASET

Dataset	Year	Location	Sensors	Size (pixel)	Resolution (m/pixel)
Farmland	2006,2007	China, Jiangsu	Hyperion	220×430×132	30
Hermiston	2004,2007	USA, Oregon	Hyperion	390×200×224	30
BayArea	2013,2015	USA, California	AVIRIS	600×500×224	20
River	2013,2013	China,Jiangsu	Hyperion	463×241×242	30

III. EXPERIMENTS

In this section, two experiments were conducted based on six pairs of HRSIs. The first experiment aimed to verify the feasibility and superiorities of the proposed D²IAR-based change detection approach by comparing it with some state-of-the-art methods. The second experiment was designed to investigate the relationship between detection accuracies and the parameters of the adaptive region extension algorithm in our proposed approach.

A. Dataset Description

Four pairs of real HRSIs were adopted in the following experiments, the parameters of datasets are shown in Table I. As shown in Fig. 4, each dataset includes bitemporal HRSIs and a ground truth image. Details of each dataset are given below.

1) Dataset-1: This dataset was named "Hermiston." Data were collected on an irrigated farmland in the Hermiston area of Oregon, USA, in 2004 and 2007 by a Hyperion sensor with a spectral range of 0.4–2.5 μm and a spatial resolution of 30 m on the EO-1 satellite. Each image with a size of 390×220 pixels includes 224 spectral bands. The change event was the differential growth of crops caused by an automatic irrigation device.

2) Dataset-2: This dataset was named "River" to refer to the change event of a delta in a river in Jiangsu Province, China. The bitemporal images were acquired on May 3 and December 31, 2013. The dataset has 242 spectral bands with a spectral range of 350–2580 nm, and its spatial resolution is 30 m/pixel with a size of 463 × 241 pixels.

3) Dataset-3: This dataset was called "Farmland" to describe a wetland area change in Yancheng City, Jiangsu province, China. The data were acquired by the EO-1 Hyperion sensor on May 3, 2006 and April 23, 2007. The dataset has 132 spectral bands and a spatial size of 420×140 pixels with 30 m/pixel spatial resolution. The change in "Farmland" was mainly caused by agricultural changes.

4) Dataset-4: This dataset was named "Bay Area," which is located at the city of Patterson in the San Francisco Bay Area, California, USA. The two HRSIs were taken in 2013 and 2015 by the AVIRIS sensor, which contains 224 spectral bands. The study area of this dataset is 600 × 500 pixels with a resolution of 20 m/pixel. Note that the gray, black, and white pixels denote the uninterested, unchanged, and changed areas.

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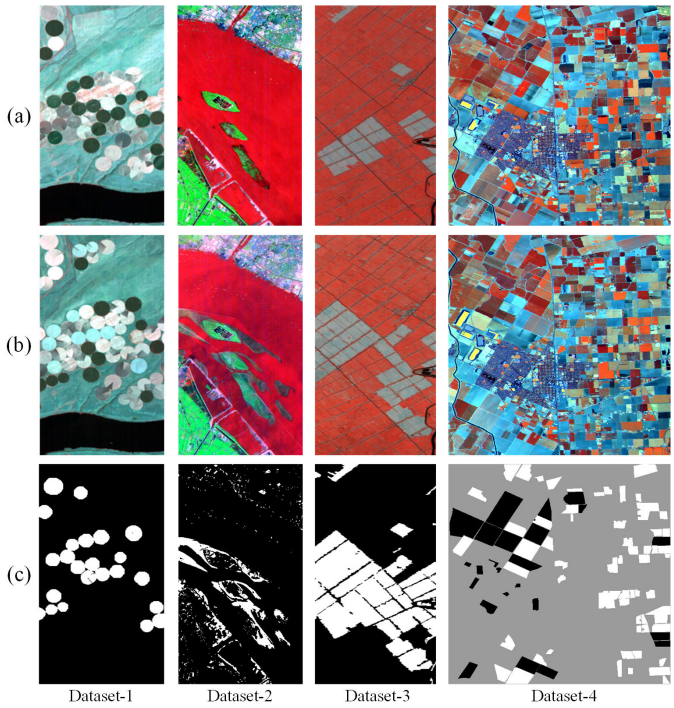


Fig. 4. Datasets for experiments. (a) Preevent image. (b) Postevent image. (c) Ground reference map.

B. Experiment Setting

In this section, eight excellent change detection algorithms were adopted to compare with our proposed approach and verify its feasibility and effectiveness. Note that the parameters of the traditional methods were used as the default in the corresponding reference, and the parameters of the deep learning approach were determined by trial-and-error experiments. The optimized parameters of each approach are briefly summarized in Table II. Details of the selected methods for comparisons are presented below.

CVA [52]: This method treats each pixel in HRSIs as a vector and measures the change magnitude by comparing the Euclidean distances between the vectors. The binary change map is then obtained by applying a binary threshold to divide the CMI.

Patch tensor change detection (PTCD) [57]: The method uses tensors to express the structure of HRSIs and combine the spatial–spectral information. PTCD aims to reflect differences by breaking down the data.

SFA [21]: This method aims to explore an invariant feature space that minimizes the difference in transformed features of the unchanged pixel so that the change area can be enhanced.

Tucker decomposition and reconstruction detector (TDRD) [58]: Tucky decomposer is adopted to decompose and reconstruct HRSIs. The decomposed tuckers are used to measure the change magnitude between the bitemporal images.

Support vector machine (SVM) [59]: In this approach, spatial–spectral features are explored to measure change magnitude, and binary change detection tasks are regarded as a binary classification problem. Thus, SVM is used instead of a binary threshold to obtain the final BCDM.

K-nearest neighbor (KNN) [60]: This approach concentrates on utilizing the contextual information around a pixel with KNN, which means that K-neighboring pixels around a pixel are adopted to smoothen the noise in a change detection map.

TABLE II
OPTIMIZED PARAMETER SETTING OF EACH APPROACH FOR EACH DATASET

Dataset	Method	Parameter
Hermiston	PTCD [57]	Patchsize=3
	TDRD [58]	η=0.99
	SVM [59]	Sample=0.2
	KNN [60]	Sample=0.2
	CSANET [61]	Sample=0.1
	GETNET [36]	Sample=0.7
River	PTCD [57]	Patchsize=3
	TDRD [58]	η=0.90
	SVM [59]	Sample=0.2
	KNN [60]	Sample=0.2
	CSANET [61]	Sample=0.03
	GETNET [36]	Sample=0.7
Farmland	PTCD [57]	Patchsize=4
	TDRD [58]	η=0.955
	SVM [59]	Sample=0.2
	KNN [60]	Sample=0.2
	CSANET [61]	Sample=0.1
	GETNET [36]	Sample=0.7
Bay Area	PTCD [57]	Patchsize=5
	TDRD [58]	η=0.99
	SVM [59]	Sample=0.2
	KNN [60]	Sample=0.2
	CSANET [61]	Sample=0.1
	GETNET [36]	Sample=0.7

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Cross-temporal interaction symmetric attention network (CSANET) [61]: In this approach, a backbone is combined with self-attention to enhance the feature representation of each temporal image. Moreover, differential features are explored from each temporal image to obtain a change magnitude map.

General End-To-End 2D-CNN Framework (GETNET) [36]: The network unmixes the pixels to obtain the pure features and then puts these features into the CNN to obtain the change detection map.

Six widely used indicators were also utilized to assess the performance of each approach for quantitative comparisons [62], including overall accuracy (OA), average accuracy (AA), kappa coefficient (KA), false alarm (FA), missing alarm (MA), and total error (TE). Details of these metrics can be tracked in [55].

C. Experimental Results

Based on the above experimental setting, the experimental results drawn from four pairs of real HRSIs are given below.

1) Results on Dataset-1: As shown in Table III, the proposed method achieved good performance in terms of OA, KA, AA, FA, and TE. The improvement was about 0.11%-4.95% comparing with using SVM [59], KNN [60], CSANET [61], and GETNET [36]. In addition, the KA = 0.9417 and TE = 1.296 achieved by our proposed approach were the best accuracies in all the methods. The visual comparison in Fig. 5 shows the change detection map obtained with the proposed method. For the CSANET [61], although it achieved good accuracy, it was still unable to accurately distinguish the edges of changing areas when some crop areas are close to one another and the boundaries of the irrigation areas are connected. On the contrary, the proposed approach can detect change regions more accurately and identify the boundaries of ground objects better.

2) Results on Dataset-2: The quantitative comparison and visual comparison results based on Dataset-2 are summarized and compared in Table IV and Fig. 6, respectively. From the comparisons, the observation indicated that the following.

1) The proposed approach achieved competitive accuracy compared with using KNN [60] and CSANET [61] in terms of OA, but the proposed approach obtained higher accuracies than other methods in terms of AA and Kappa.

2) The visual observation demonstrated that the detection map obtained with the proposed approach performed with less missed detection pixels (red color) and false detection pixels (aqua color).

3) Results on Dataset-3: Based on the quantitative comparison in Table V, the proposed calculation method achieved competitive accuracies in terms of OA, KA, AA, and TE. The corresponding visual observations in Fig. 7 can support the conclusion from the quantitative comparisons. For example, we can see that the detection maps obtained by CVA [52], SFA [21], KNN [60], and GETNET [36] performed with more salt-and-pepper noise than those acquired with the proposed approach. Although SVM [59] and TDRD [58] achieved satisfactory accuracy and avoided noise in the detection maps, the boundaries among different changed areas are false detection. CSANET [61] and our proposed approach obtained similar accuracies in terms of OA, TE, and KA. However, the proposed method was superior in terms of missed detection of changed pixels.

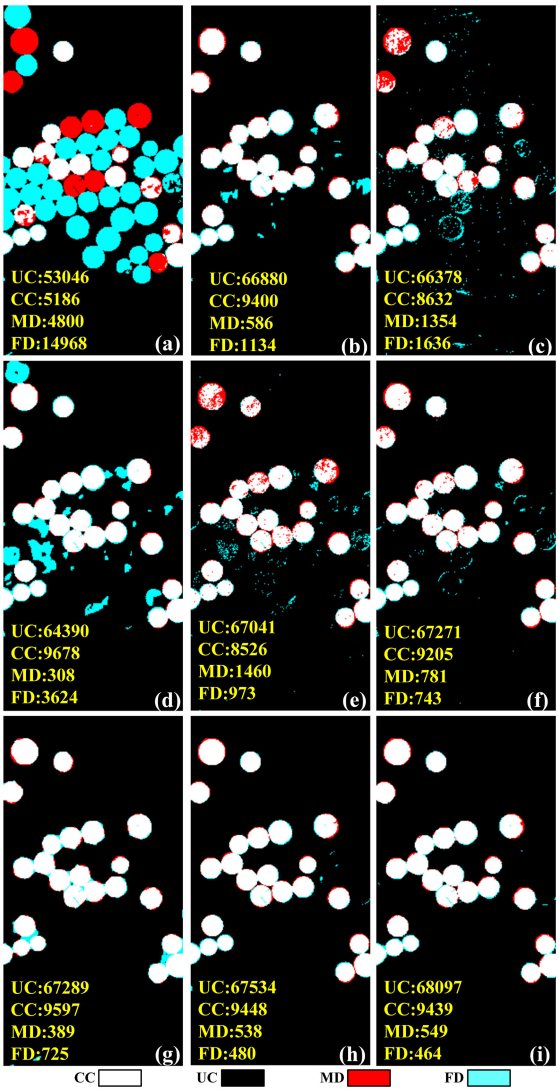


Fig. 5. Acquired change detection map using different methods for Dataset-1. (a) CVA [52]. (b) PTCD [57]. (c) SFA [21]. (d) TDRD [58]. (e) SVM [59]. (f) KNN [60]. (g) CSANET [61]. (h) GETNET [36]. (i) Proposed D²IAR-based change detection approach.

4) Results on Dataset-4: For the Bay Area datasets, the quantitative comparisons are presented in Table VI, and all the approaches achieved high OAs larger than 90%, with the proposed approach achieving the top OA of 97.92%. Moreover, the proposed approach achieved the five best accuracies in terms of the six evaluation indicators. As shown in Fig. 8, the detection map achieved by our proposed approach had less noise and less missed detection and false detection pixels compared with the maps achieved by other approaches.

D. Discussion

To further extend the application of the proposed approach, the relationship between the detection accuracies and the

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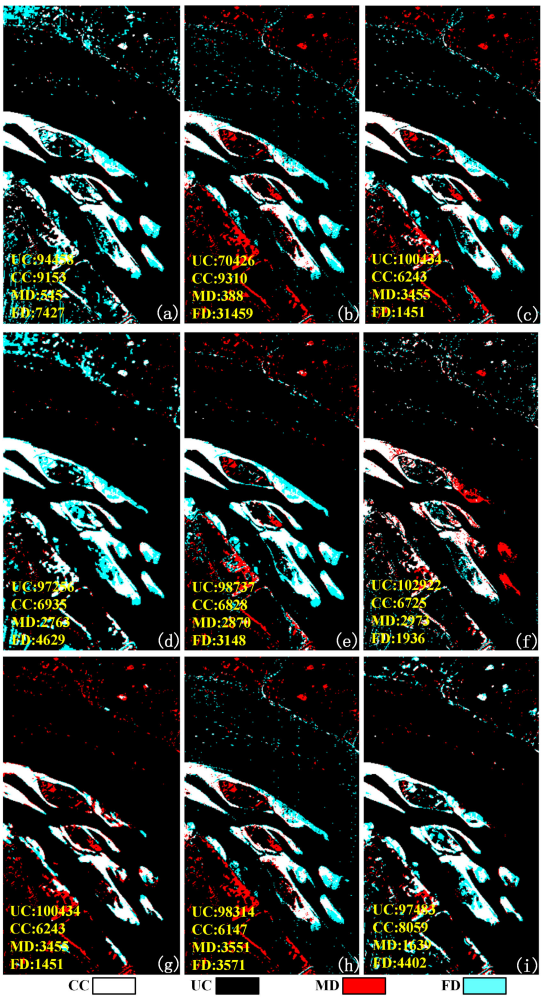


Fig. 6. Acquired change detection map using different methods for Dataset-2. (a) CVA [52]. (b) PTCD [57]. (c) SFA [21]. (d) TDRD [58]. (e) SVM [59]. (f) KNN [60]. (g) CSANET [61]. (h) GETNET [36]. (i) Proposed D^{2}IAR-based change detection approach.

TABLE III

QUANTITATIVE COMPARISON BETWEEN THE SELECTED METHODS AND THE PROPOSED D^{2}IAR-BASED CHANGE DETECTION APPROACH FOR DATASET-1

Method	OA	AA	Kappa	FA	MA	TE
CVA [52]	74.66	64.96	0.2086	22.01	48.07	25.34
PTCD [57]	97.79	96.23	0.9035	1.667	5.868	2.205
SFA [21]	96.17	92.02	0.8304	2.405	13.56	3.833
TDRD [58]	94.96	95.79	0.8022	5.328	**3.084**	5.041
SVM [59]	96.88	91.97	0.8573	1.431	14.62	3.119
KNN [60]	98.05	95.54	0.9123	1.092	7.821	1.954
CSANET [61]	98.57	96.52	0.9369	1.066	3.895	1.428
GETNET [36]	98.05	95.54	0.9123	1.092	7.821	1.954
Proposed method	**98.70**	**96.92**	**0.9417**	**0.6822**	5.478	**1.296**

parameters (T_1 and T_2) of the proposed approach are compared. Detailed discussions are presented below.

1) Detection accuracies versus the change of T_1: As shown in Fig. 9, the accuracies of OA, KA, and AA decreased gradually when the setting of T_1 was increased from 5 to 100 for Dataset-1, and a similar phenomenon was reflected in Dataset-4. For Dataset-2, the accuracies of OA, KA, and AA increased when T_1 was increased from 5 to 25, and then

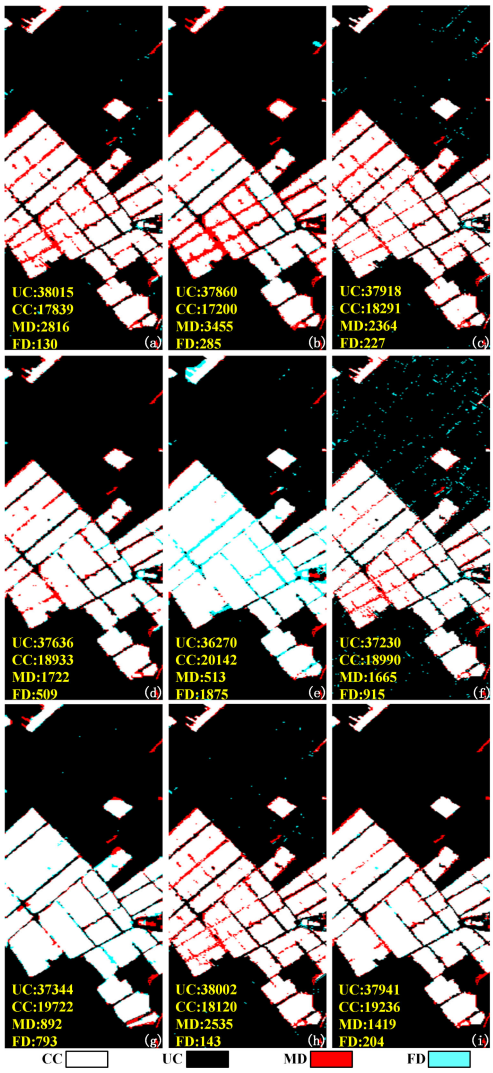


Fig. 7. Acquired change detection map using different methods for Dataset-3. (a) CVA [52]. (b) PTCD [57]. (c) SFA [21]. (d) TDRD [58]. (e) SVM [59]. (f) KNN [60]. (g) CSANET [61]. (h) GETNET [36]. (i) Proposed D^{2}IAR-based change detection approach.

TABLE IV

QUANTITATIVE COMPARISON BETWEEN THE SELECTED METHODS AND THE PROPOSED D^{2}IAR-BASED CHANGE DETECTION APPROACH FOR DATASET-2

Method	OA	AA	Kappa	FA	MA	TE
CVA [52]	92.86	90.55	0.6593	7.29	**5.62**	7.144
PTCD [57]	71.46	82.56	0.2659	30.88	4.001	28.54
SFA [21]	94.61	83.66	0.6645	3.09	29.59	5.393
TDRD [58]	92.37	91.19	0.637	7.592	8.033	7.63
SVM [59]	93.38	83.48	0.616	4.543	28.49	6.625
KNN [60]	**95.60**	83.72	0.7087	**1.90**	30.66	4.399
CSANET [61]	**95.60**	81.47	0.6844	1.424	35.63	**4.397**
GETNET [36]	93.62	79.94	0.5982	6.505	36.62	6.383
Proposed method	95.10	**91.41**	**0.7285**	4.122	13.05	4.899

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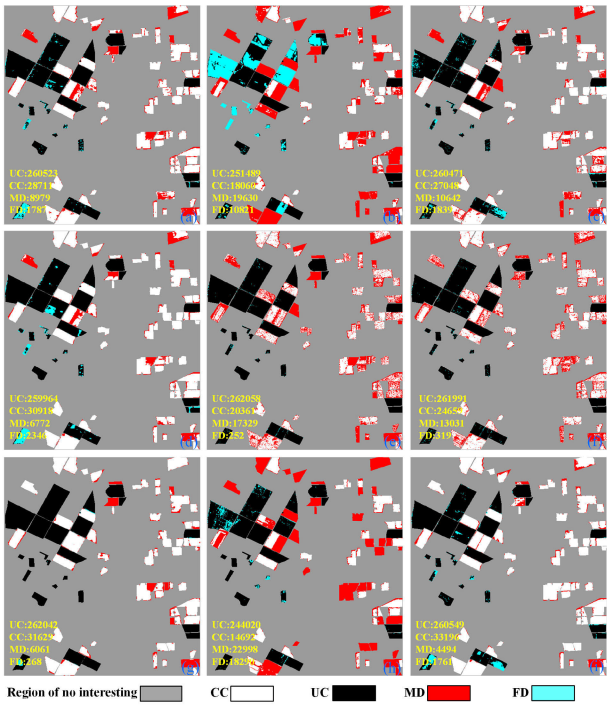


Fig. 8. Acquired change detection map using different methods for Dataset-4. (a) CVA [52]. (b) PTCD [57]. (c) SFA [21]. (d) TDRD [58]. (e) SVM [59]. (f) KNN [60]. (g) CSANET [61]. (h) GETNET [36]. (i) Proposed D²IAR-based change detection approach.

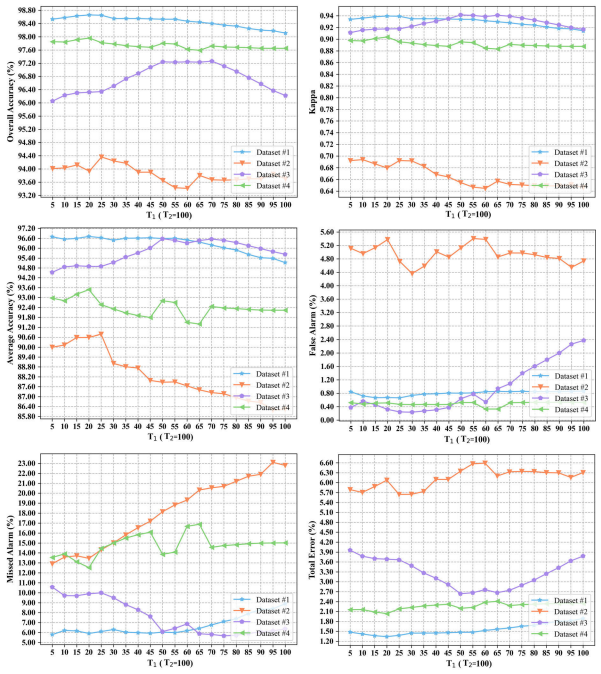


Fig. 9. Relationship between detection accuracies and the parameter T_1 of the proposed approach (T_2 is fixed at 100).

the accuracies tended to decrease subsequently. Therefore, the optimized compositions of T_1 and T_2 are different for a specific dataset. For example, the optimized T_1 and T_2 are 20 and 85 for Dataset-1 in terms of OA, but $T_1 = 25$ and $T_2 = 20$ are the optimized compositions for Dataset-2 in terms of OA.

2) Detection accuracies versus the change of T_2: Based on the observation of the relationship between T_1 and the detection accuracies, T_1 was fixed at a specific value for each

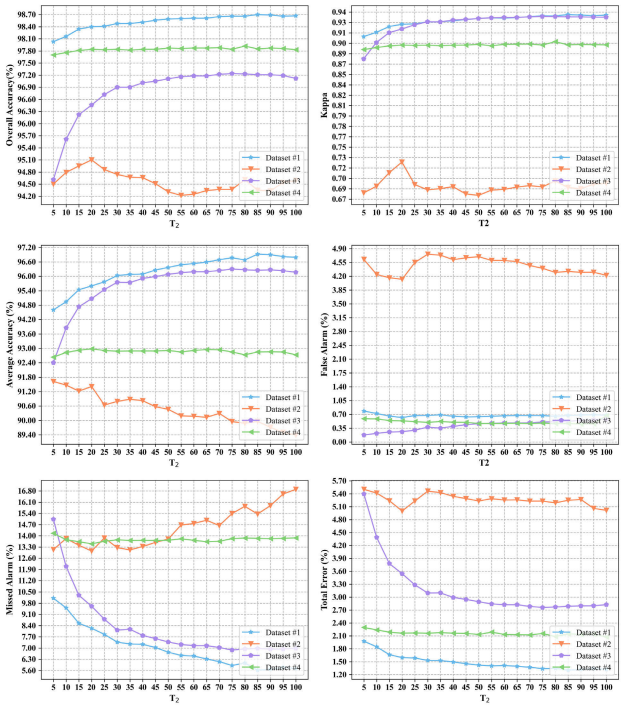


Fig. 10. Relationship between detection accuracies and the parameter T_2 of the proposed approach (T_1 is fixed at 20, 25, 60, and 20 for Dataset-1, Dataset−2, Dataset−3, and Dataset−4, respectively).

TABLE V

QUANTITATIVE COMPARISON BETWEEN THE SELECTED METHODS AND THE PROPOSED D²IAR-BASED CHANGE DETECTION APPROACH FOR DATASET-3

Method	OA	AA	Kappa	FA	MA	TE
CVA [52]	94.99	93.01	0.8867	**0.3408**	13.63	5.01
PTCD [57]	93.64	91.26	0.8554	0.7471	16.73	6.361
SFA [21]	95.59	93.98	0.9010	0.5951	11.45	4.406
TDRD [58]	96.21	95.16	0.9156	1.334	8.337	3.794
SVM [59]	95.94	96.3	0.9122	4.915	**2.484**	4.061
KNN [60]	95.61	94.77	0.9029	2.399	8.061	4.388
CSANET [61]	97.10	96.21	0.9381	2.079	4.182	2.815
GETNET [36]	95.45	93.68	0.8973	0.3749	12.27	4.554
Proposed method	**97.36**	**96.57**	**0.9415**	0.7707	6.091	**2.639**

dataset. The detailed settings of T_1 are shown in Fig. 10. From the observation of Fig. 10, the OAs of the proposed approach remained at 98.6 and 97.8, respectively, when T_2 was increased from 5 to 100 for Dataset-1 and Dataset-4. The observation also revealed that Dataset-2 (River) was sensitive to the change of T_2. This may be because the change scene is located at the zone of water and land intersection, and so the change scene is complex in bitemporal HRSIs.

Based on the discussion, the different datasets perform different sensitivities to the change of T_1 and T_2 in the applications of the proposed approach. Therefore, optimized compositions of T_1 and T_2 should be given via trial-and-error experiments for a specific dataset when applying the proposed approach.

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TABLE VI
QUANTITATIVE COMPARISON BETWEEN THE SELECTED METHODS AND THE PROPOSED D^{2}IAR-BASED CHANGE DETECTION APPROACH FOR DATASET-4

Method	OA	AA	Kappa	FA	MA	TE
CVA [52]	96.41	87.75	0.8221	0.6813	23.82	3.589
PTCD [57]	93.13	73.77	0.3744	0.3744	52.08	6.871
SFA [21]	95.84	85.53	0.7896	0.7011	28.24	4.16
TDRD [58]	96.96	90.57	0.8543	0.8944	17.97	3.039
SVM [59]	94.14	76.96	0.6691	0.09607	45.98	5.86
KNN [60]	95.55	82.65	0.7633	0.1216	34.57	4.45
CSANET [61]	97.89	91.91	0.8972	**0.1022**	16.08	2.11
GETNET [36]	91.84	69.21	0.5085	0.5638	61.02	8.159
Proposed method	**97.92**	**93.73**	**0.9022**	0.7707	**6.091**	**2.083**

IV. CONCLUSION

In this article, a novel D^{2}IAR-based change detection approach was proposed for LCCD with HRSIs. In the proposed D^{2}IAR-based change detection approach, contextual information around each pixel was explored by generating an adaptive region around each pixel. Then, KDE and NSST were introduced to define a metric distance based on spectral value distribution within each pairwise adaptive region and fuse the CMIs from each pairwise band into the final CMI. The applications of the proposed D^{2}IAR-based change detection approach on the four pairs of real HRSIs well indicated the feasibility of the proposed approach, and the comparison with eight state-of-art methods demonstrated the superiorities of the proposed D^{2}IAR-based change detection approach.

The experiments and comparisons verified the feasibility and superiorities of the proposed D^{2}IAR-based change detection approach for the tasks of LCCD with HRSIs. However, two aspects of the proposed approach can be improved in future studies.

1) For T_1 and T_2, the adaptive region extension algorithm used in our proposed approach should be estimated automatically because the determination of the optimized T_1 and T_2 for a dataset are time-consuming.

2) The datasets used in the experiments are small, and so larger datasets should be adopted to test the robustness of the proposed approach in future studies.

REFERENCES

[1] Z. Lv et al., "Land cover change detection with heterogeneous remote sensing images: Review, progress, and perspective," *Proc. IEEE*, vol. 110, no. 12, pp. 1976–1991, Dec. 2022.

[2] Z. Lv, T. Liu, J. A. Benediktsson, and N. Falco, "Land cover change detection techniques: Very-high-resolution optical images: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 44–63, Mar. 2022.

[3] W. Shi, M. Zhang, R. Zhang, S. Chen, and Z. Zhan, "Change detection based on artificial intelligence: State-of-the-art and challenges," *Remote Sens.*, vol. 12, no. 10, p. 1688, May 2020.

[4] Z. Li, W. Shi, P. Lu, L. Yan, Q. Wang, and Z. Miao, "Landslide mapping from aerial photographs using change detection-based Markov random field," *Remote Sens. Environ.*, vol. 187, pp. 76–90, Dec. 2016.

[5] H. Yin, D. Pflugmacher, A. Li, Z. Li, and P. Hostert, "Land use and land cover change in inner Mongolia—Understanding the effects of China's re-vegetation programs," *Remote Sens. Environ.*, vol. 204, pp. 918–930, Jan. 2018.

[6] P. Coppin, I. Jonckheere, K. Nackaerts, B. Muys, and E. Lambin, "Review ArticleDigital change detection methods in ecosystem monitoring: A review," *Int. J. Remote Sens.*, vol. 25, no. 9, pp. 1565–1596, May 2004.

[7] C. Wu, B. Du, and L. Zhang, "A subspace-based change detection method for hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 815–830, Apr. 2013.

[8] S. Liu, D. Marinelli, L. Bruzzone, and F. Bovolo, "A review of change detection in multitemporal hyperspectral images: Current techniques, applications, and challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 140–158, Jun. 2019.

[9] M. T. Eismann and J. Meola, "Hyperspectral change detection: Methodology and challenges," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, vol. 2, Jul. 2008, p. 605.

[10] Z. Y. Lv, T. F. Liu, P. Zhang, J. A. Benediktsson, T. Lei, and X. Zhang, "Novel adaptive histogram trend similarity approach for land cover change detection by using bitemporal very-high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 9554–9574, Dec. 2019.

[11] Z. Lv, P. Zhong, W. Wang, Z. You, J. A. Benediktsson, and C. Shi, "Novel piecewise distance based on adaptive region key-points extraction for LCCD with VHR remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5607709.

[12] J. Qu, S. Hou, W. Dong, Y. Li, and W. Xie, "A multilevel encoder–decoder attention network for change detection in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518113.

[13] J. Qu, Y. Xu, W. Dong, Y. Li, and Q. Du, "Dual-branch difference amplification graph convolutional network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5519912.

[14] F. Bovolo, S. Marchesi, and L. Bruzzone, "A framework for automatic and unsupervised detection of multiple changes in multitemporal images," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 6, pp. 2196–2212, Jun. 2012.

[15] S. Liu, L. Bruzzone, F. Bovolo, M. Zanetti, and P. Du, "Sequential spectral change vector analysis for iteratively discovering and detecting multiple changes in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 8, pp. 4363–4378, Aug. 2015.

[16] F. Thonfeld, H. Feilhauer, M. Braun, and G. Menz, "Robust change vector analysis (RCVA) for multi-sensor very high resolution optical satellite data," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 50, pp. 131–140, Aug. 2016.

[17] X. Ou, L. Liu, S. Tan, G. Zhang, W. Li, and B. Tu, "A hyperspectral image change detection framework with self-supervised contrastive learning pretrained model," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 7724–7740, 2022.

[18] M. Sohail, Z. Chen, and G. Liu, "Tensor ring with alternative change mask for multitemporal hyperspectral image change detection," *Pattern Recognit. Lett.*, vol. 164, pp. 46–52, Dec. 2022.

[19] A. A. Nielsen, K. Conradsen, and J. J. Simpson, "Multivariate alteration detection (MAD) and MAF postprocessing in multispectral, bitemporal image data: New approaches to change detection studies," *Remote Sens. Environ.*, vol. 64, no. 1, pp. 1–19, 1998.

[20] A. A. Nielsen and M. J. Canty, "Multi- and hyperspectral remote sensing change detection with generalized difference images by the IR-MAD method," in *Proc. Int. Workshop Analysis Multi-Temporal Remote Sens. Images*, 2005, pp. 169–173.

[21] C. Wu, B. Du, and L. Zhang, "Slow feature analysis for hyperspectral change detection," in *Proc. 6th Workshop Hyperspectral Image Signal Process., Evol. Remote Sens. (WHISPERS)*, Jun. 2014, pp. 1–4.

[22] V. Ortiz-Rivera, M. Velez-Reyes, and B. Roysam, "Change detection in hyperspectral imagery using temporal principal components," *Proc. SPIE*, vol. 6233, pp. 368–377, May 2006.

[23] J. S. Deng, K. Wang, Y. H. Deng, and G. J. Qi, "PCA-based land-use change detection and analysis using multitemporal and multisensor satellite data," *Int. J. Remote Sens.*, vol. 29, no. 16, pp. 4823–4838, Aug. 2008.

[24] J. Zhou, C. Kwan, B. Ayhan, and M. T. Eismann, "A novel cluster kernel RX algorithm for anomaly and change detection using hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 11, pp. 6497–6504, Nov. 2016.

[25] B. Demir, F. Bovolo, and L. Bruzzone, "Detection of land-cover transitions in multitemporal remote sensing images with active-learning-based compound classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 5, pp. 1930–1941, May 2012.

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[26] C. Wu, B. Du, X. Cui, and L. Zhang, "A post-classification change detection method based on iterative slow feature analysis and Bayesian soft fusion," *Remote Sens. Environ.*, vol. 199, pp. 241–255, Sep. 2017.

[27] Z. Hou and W. Li, "A joint morphological profiles and patch tensor change detection for hyperspectral imagery," 2022, *arXiv:2201.08027*.

[28] T. Zhan, Y. Sun, Y. Tang, Y. Xu, and Z. Wu, "Tensor regression and image fusion-based change detection using hyperspectral and multispectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 9794–9802, 2021.

[29] Z. Lv, P. Zhang, W. Sun, J. Atli Benediktsson, and T. Lei, "Novel land-cover classification approach with nonparametric sample augmentation for hyperspectral remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4407613.

[30] A. Torfi, R. A. Shirvani, Y. Keneshloo, N. Tavaf, and E. A. Fox, "Natural language processing advancements by deep learning: A survey," 2020, *arXiv:2003.01200*.

[31] N. O'Mahony et al., "Deep learning vs. traditional computer vision," in *Advances in Computer Vision*, vol. 1. Berlin, Germany: Springer, 2020, pp. 128–144.

[32] L. Khelifi and M. Mignotte, "Deep learning for change detection in remote sensing images: Comprehensive review and meta-analysis," *IEEE Access*, vol. 8, pp. 126385–126400, 2020.

[33] J. Shi, Z. Zhang, T. Wu, X. Zhang, and Y. Lei, "Collaborative self-perception network architecture for hyperspectral image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[34] Y. Yang, J. Qu, S. Xiao, W. Dong, Y. Li, and Q. Du, "A deep multiscale pyramid network enhanced with spatial–spectral residual attention for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5525513.

[35] Y. Zhang et al., "Depthwise separable convolutional autoencoders for hyperspectral image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[36] Q. Wang, Z. Yuan, Q. Du, and X. Li, "GETNET: A general end-to-end 2-D CNN framework for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 3–13, Jan. 2018.

[37] T. Zhan et al., "TDSSC: A three-directions spectral–spatial convolution neural network for hyperspectral image change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 377–388, 2020.

[38] M. Hu, C. Wu, and L. Zhang, "HyperNet: Self-supervised hyperspectral spatial–spectral feature understanding network for hyperspectral change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5543017.

[39] L. Mou, L. Bruzzone, and X. X. Zhu, "Learning spectral–spatial–temporal features via a recurrent convolutional neural network for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 924–935, Feb. 2019.

[40] Z. Lv, P. Zhang, W. Sun, T. Lei, J. Atli Benediktsson, and P. Li, "Sample iterative enhancement approach for improving classification performance of hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[41] M. Gong, L. Su, M. Jia, and W. Chen, "Fuzzy clustering with a modified MRF energy function for change detection in synthetic aperture radar images," *IEEE Trans. Fuzzy Syst.*, vol. 22, no. 1, pp. 98–109, Feb. 2014.

[42] T. Celik, "Unsupervised change detection in satellite images using principal component analysis and k-means clustering," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 4, pp. 772–776, Oct. 2009.

[43] J. Lopez-Fandino, A. S. Garea, D. B. Heras, and F. Arguello, "Stacked autoencoders for multiclass change detection in hyperspectral images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 1906–1909.

[44] A. Song, J. Choi, Y. Han, and Y. Kim, "Change detection in hyperspectral images using recurrent 3D fully convolutional networks," *Remote Sens.*, vol. 10, no. 11, p. 1827, Nov. 2018.

[45] F. Luo, T. Zhou, J. Liu, T. Guo, X. Gong, and J. Ren, "Multiscale diff-changed feature fusion network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502713.

[46] Z. Lv, H. Huang, W. Sun, T. Lei, J. Atli Benediktsson, and J. Li, "Novel enhanced UNet for change detection using multimodal remote sensing image," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[47] H. Chen, C. Wu, B. Du, and L. Zhang, "Deep Siamese multi-scale convolutional network for change detection in multi-temporal VHR images," in *Proc. 10th Int. Workshop Anal. Multitemporal Remote Sens. Images (MultiTemp)*, Aug. 2019, pp. 1–4.

[48] D. Lu, S. Cheng, L. Wang, and S. Song, "Multi-scale feature progressive fusion network for remote sensing image change detection," *Sci. Rep.*, vol. 12, no. 1, p. 11968, Jul. 2022.

[49] D. Lu, P. Mausel, E. Brondizio, and E. Moran, "Change detection techniques," *Int. J. Remote Sens.*, vol. 25, no. 12, pp. 2365–2401, Apr. 2004.

[50] Z. Lv, J. Liu, W. Sun, T. Lei, J. A. Benediktsson, and X. Jia, "Hierarchical attention feature fusion-based network for land cover change detection with homogeneous and heterogeneous remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4411115.

[51] G. Easley, D. Labate, and W.-Q. Lim, "Sparse directional image representations using the discrete shearlet transform," *Appl. Comput. Harmon. Anal.*, vol. 25, no. 1, pp. 25–46, Jul. 2008.

[52] F. Bovolo and L. Bruzzone, "A theoretical framework for unsupervised change detection based on change vector analysis in the polar domain," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 1, pp. 218–236, Jan. 2006.

[53] Q. Wang, F. Zhang, and X. Li, "Hyperspectral band selection via optimal neighborhood reconstruction," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8465–8476, Dec. 2020.

[54] Z. Lv, M. Zhang, W. Sun, J. Atli Benediktsson, T. Lei, and N. Falco, "Spatial-contextual information utilization framework for land cover change detection with hyperspectral remote sensed images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4411911.

[55] Z. Lv, F. Wang, T. Liu, X. Kong, and J. A. Benediktsson, "Novel automatic approach for land cover change detection by using VHR remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[56] C. Benedek, M. Shadaydeh, Z. Kato, T. Szirányi, and J. Zerubia, "Multilayer Markov random field models for change detection in optical remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 107, pp. 22–37, Sep. 2015.

[57] Z. Hou, W. Li, and Q. Du, "A patch tensor-based change detection method for hyperspectral images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2021, pp. 4328–4331.

[58] Z. Hou, W. Li, R. Tao, and Q. Du, "Three-order Tucker decomposition and reconstruction detector for unsupervised hyperspectral change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 6194–6205, 2021.

[59] F. Bovolo, L. Bruzzone, and M. Marconcini, "A novel approach to unsupervised change detection based on a semisupervised SVM and a similarity measure," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 7, pp. 2070–2082, Jul. 2008.

[60] T. Abeywickrama, M. A. Cheema, and D. Taniar, "K-nearest neighbors on road networks: A journey in experimentation and in-memory implementation," 2016, *arXiv:1601.01549*.

[61] R. Song, W. Ni, W. Cheng, and X. Wang, "CSANet: Cross-temporal interaction symmetric attention network for hyperspectral image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

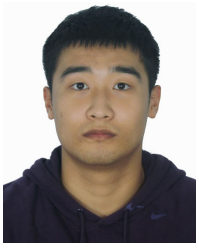
[62] B. Krishnamurthy, S. Sen, Y. Zhang, and Y. Chen, "Sketch-based change detection: Methods, evaluation, and applications," in *Proc. SIGCOMM*, 2003, pp. 234–247.



Zhiyong Lv (Senior Member, IEEE) received the M.S. and Ph.D. degrees from the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China, in 2008 and 2014, respectively.

He was an Engineer of surveying and worked in The First Institute of Photogrammetry and Remote Sensing, Xi'an, China, from 2008 to 2011. He is currently working with the School of Computer Science and Engineering, Xi'an University of Technology, Xi'an. His research interests include multi hyperspectral and high-resolution remotely sensed image processing, spatial feature extraction, neural networks, pattern recognition, deep learning, and remote sensing applications.

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Zhengjie Lei received the B.S. degree from the Department of Mathematics, Xi'an University of Technology, Xi'an, China, in 2023, where he is currently pursuing the master's degree in computer science.

He is interested in hyperspectral remote sensing images change detection, pattern recognition, and deep-learning.



Linfu Xie received the bachelor's and Ph.D. degrees from Wuhan University, Wuhan, Hubei, China, in 2013 and 2019, respectively.

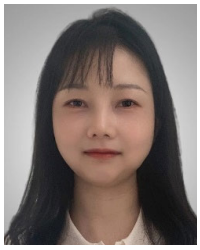
He joined the School of Architecture and Urban Planning, Shenzhen University, Shenzhen, China, in 2019, where he is currently an Assistant Professor. His research interests include image processing, point clouds processing, multi-platform photogrammetry, and 3-D city modeling.



Nicola Falco (Senior Member, IEEE) received the B.Sc. and M.Sc. degrees in telecommunication engineering from the University of Trento, Trento, Italy, in 2008 and 2011, respectively, and the Joint Ph.D. degree in electrical and computer engineering from the University of Iceland, Reykjavik, Iceland, and in information and communication technologies from the University of Trento, in 2015.

He is currently a Research Scientist with the Lawrence Berkeley National Laboratory, Climate and Ecosystem Sciences Division, Berkeley, CA, USA. His research focuses on the development of methodologies based on image/signal processing, pattern recognition, and machine learning for multi-source data analysis and integration, including remote sensing (hyperspectral, LiDAR, high-res) and geophysical data, with applications in environmental monitoring, climate change, and precision agriculture.

Dr. Falco was a recipient of the Recognition of IEEE Geoscience Remote Sensing Letters Best Reviewers in 2013, and a co-recipient of the third price in the Student Paper Competition of the 2015 IEEE International Geoscience and Remote Sensing Symposium. He is a referee for several international journals, including IEEE TRANSACTION ON GEOSCIENCE AND REMOTE SENSING, IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, IEEE JOURNAL OF SELECTED TOPIC IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, *International Journal of Applied Earth Observation and Geoinformation* (Elsevier), *Pattern Recognition Letters* (Elsevier), *International Journal of Remote Sensing*, textitRemote Sensing Letters (Taylor & Francis), *Agriculture* (MDPI), *Remote Sensing* (MDPI), and *Scientific Reports* (Nature).



Cheng Shi (Member, IEEE) received the B.S. degree from the Xi'an University of Architecture and Technology, Xi'an, Shanxi, China, in 2009, and the M.S. and Ph.D. degrees from Xidian University, Xi'an, in 2012 and 2016, respectively.

She worked as a Post-Doctoral Research Fellow with the University of Macau, Macau, China, from 2016 to 2018. She is currently working as an Associate Professor with the School of Computer Science and Engineering, Xi'an University of Technology, Xi'an. Her research interests include deep learning, remote-sensing image classification, and pattern recognition.



Zhenzhen You (Member, IEEE) received the B.S. degree in telecommunication from Xidian University, Xi'an, China, in 2011, the M.Eng. and M.Res. degrees in image processing from IMT Atlantique (ex Télécom Bretagne), Brest, France, in 2014, and the Ph.D. degree in computer science from Pierre and Marie Curie University (University of Paris VI), Paris, France, in 2017.

From 2014 to 2017, she did research at the Molecular Imaging Research Center (MIRCen), CEA, Paris. Since 2018, she has been a Lecturer with the Department of Computer Science and Engineering, Xi'an University of Technology, Xi'an. Her research interests include biomedical image processing, image segmentation, image enhancement, and machine learning.

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